

30-Year Analysis of Housing Price Trends in the United States

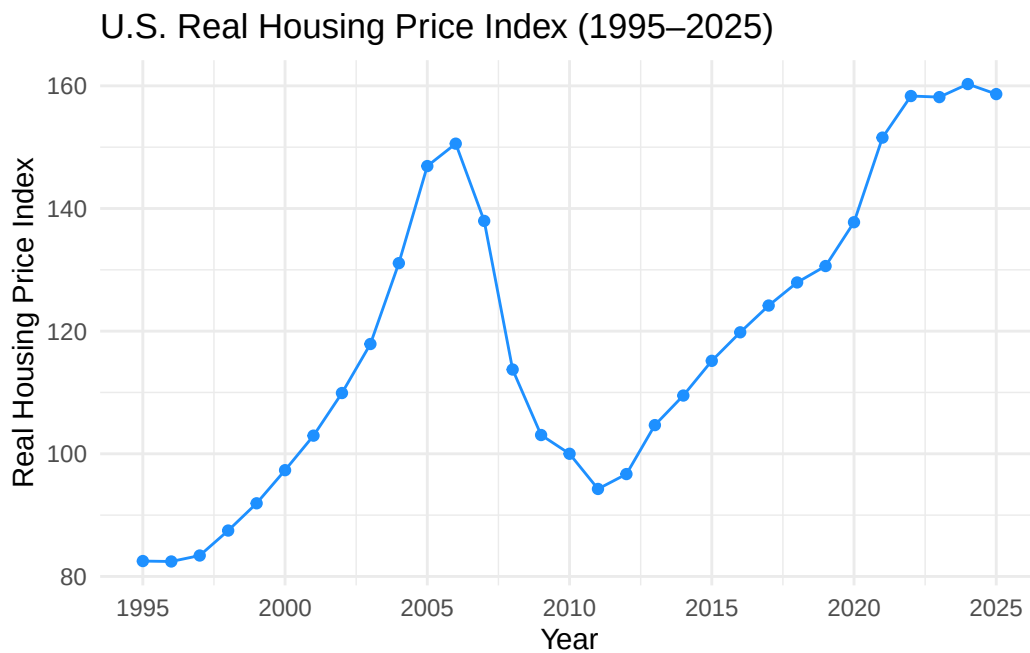
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U.S. Real Housing Price Index

Code for Price Index

Warning: package 'readr' was built under R version 4.5.3

```
[1] "year"          "Real_Housing_Index"
```



	year	Real_Housing_Index
26	1995	82.50590
27	1996	82.43685
28	1997	83.41053
29	1998	87.48183
30	1999	91.92583
31	2000	97.32615
32	2001	102.95905
33	2002	109.90865
34	2003	117.90240
35	2004	131.08735
36	2005	146.91815
37	2006	150.57413
38	2007	137.97272
39	2008	113.73963
40	2009	103.05962
41	2010	100.00397
42	2011	94.26445
43	2012	96.69220
44	2013	104.67750
45	2014	109.49972
46	2015	115.15705
47	2016	119.81035
48	2017	124.17410
49	2018	127.93960
50	2019	130.61593
51	2020	137.76130
52	2021	151.55992
53	2022	158.33925
54	2023	158.16620
55	2024	160.29122
56	2025	158.65657

Loading required package: knitr

Warning: package 'knitr' was built under R version 4.5.3

Table 1: Average Annual Real Housing Price Index (1995–2025)

	Year	Real Housing Price Index
26	1995	82.5
27	1996	82.4
28	1997	83.4
29	1998	87.5
30	1999	91.9
31	2000	97.3
32	2001	103.0
33	2002	109.9
34	2003	117.9
35	2004	131.1
36	2005	146.9
37	2006	150.6
38	2007	138.0
39	2008	113.7
40	2009	103.1
41	2010	100.0
42	2011	94.3
43	2012	96.7
44	2013	104.7
45	2014	109.5
46	2015	115.2
47	2016	119.8
48	2017	124.2
49	2018	127.9
50	2019	130.6
51	2020	137.8
52	2021	151.6
53	2022	158.3
54	2023	158.2
55	2024	160.3
56	2025	158.7

Narrative Text

Introduction

Housing prices are an important indicator of economic conditions in the United States. Changes in housing prices can reflect periods of economic growth, financial instability, and

recovery. This analysis focuses on the U.S. Real Housing Price Index from 1995 to 2025 to better understand long-term trends and major shifts in the housing market.

Data

The dataset used in this analysis comes from the Federal Reserve Economic Data (FRED) database, specifically the Real Housing Price Index (QUSR628BIS). This dataset provides monthly values that measure changes in housing prices adjusted for inflation. Using real values allows for a more accurate comparison over time because it removes the effects of inflation.

Data Preparation

To prepare the data for analysis, the monthly housing price values were grouped by year and averaged to create a single value per year. The year was extracted from the date variable and converted into a numeric format for easier plotting. The dataset was then filtered to include only the years from 1995 through 2025. Finally, the housing price index values were rounded to one decimal place to improve readability in the data table.

Visualization

The visualization is a line graph that shows the trend of the Real Housing Price Index over time. The graph reveals a steady increase in housing prices from the late 1990s through the mid-2000s, followed by a sharp decline during the 2008 financial crisis. After reaching a low point around 2011, housing prices gradually recovered and continued to rise, reaching their highest levels in the early 2020s. This pattern highlights the impact of major economic events on the housing market.

Data Table

The data table presents the average annual Real Housing Price Index values from 1995 to 2025. By summarizing the data into yearly averages, the table provides a clear and organized view of how housing prices changed over time. It allows for easy comparison between years and supports the trends observed in the visualization.

Conclusion

Overall, the Real Housing Price Index shows significant growth over the past three decades, with a major disruption during the 2008 financial crisis. The recovery period that followed demonstrates the resilience of the housing market, as prices eventually surpassed previous highs. This analysis emphasizes how economic events can strongly influence housing prices and highlights the importance of examining long-term trends when evaluating the housing market.

U.S. Inflation Rates

Code for Inflation

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
[1] "observation_date" "FPCPITOTLZGUSA"
```

```
# A tibble: 6 x 2
  observation_date FPCPITOTLZGUSA
  <date>          <dbl>
1 1960-01-01      1.46
2 1961-01-01      1.07
3 1962-01-01      1.20
4 1963-01-01      1.24
5 1964-01-01      1.28
6 1965-01-01      1.59
```

```
# A tibble: 6 x 2
  year Inflation
  <dbl>    <dbl>
1 1995     2.81
```

2	1996	2.93
3	1997	2.34
4	1998	1.55
5	1999	2.19
6	2000	3.38

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

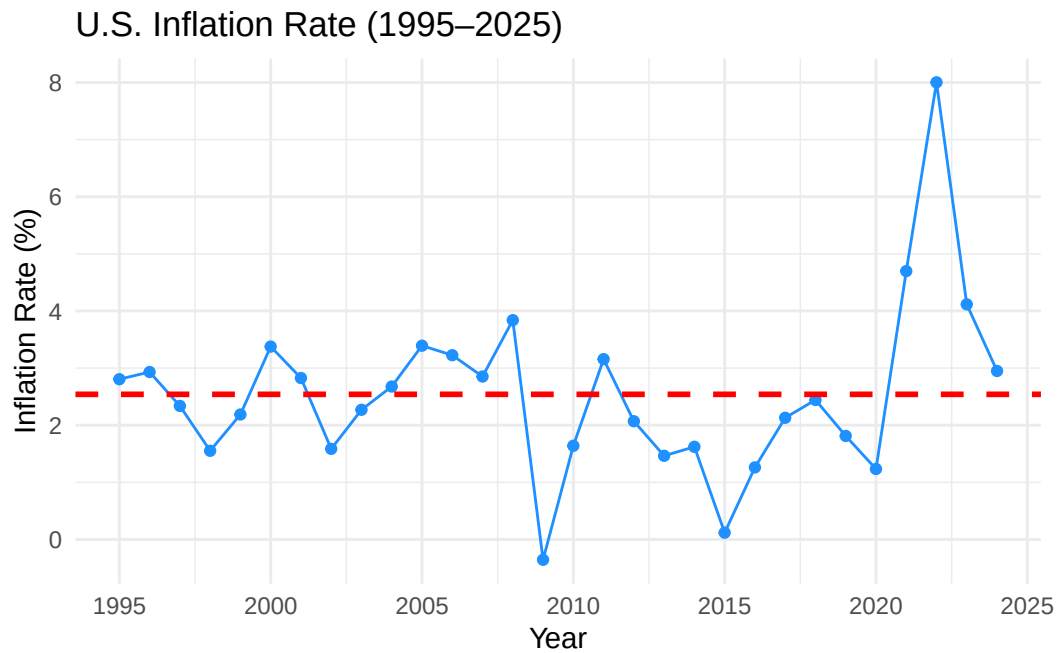


Table 2: Annual U.S. Inflation Rate (1995–2025)

Year	Inflation Rate (%)
1995	2.81
1996	2.93
1997	2.34
1998	1.55
1999	2.19
2000	3.38
2001	2.83
2002	1.59
2003	2.27
2004	2.68

Year	Inflation Rate (%)
2005	3.39
2006	3.23
2007	2.85
2008	3.84
2009	-0.36
2010	1.64
2011	3.16
2012	2.07
2013	1.46
2014	1.62
2015	0.12
2016	1.26
2017	2.13
2018	2.44
2019	1.81
2020	1.23
2021	4.70
2022	8.00
2023	4.12
2024	2.95

Narrative Text

Introduction

Inflation measures how much prices for goods and services increase over time. It is an important indicator of the overall health of the economy because it affects purchasing power and cost of living. In this analysis, I examine how the U.S. inflation rate has changed from 1995 to 2025 to better understand long-term trends and major fluctuations.

Data

The dataset used in this analysis comes from a CSV file containing U.S. inflation rates. I first load the necessary libraries for data manipulation and visualization, including `readr`, `ggplot2`, `dplyr`, and `knitr`. Then, I import the dataset and preview its structure to understand the variables included.

Data Preparation

To make the dataset easier to work with, I rename the columns to more readable names. The date variable is converted into a numeric year format so it can be used properly in the analysis. I then filtered the dataset to only include data from 1995 to 2025, which keeps the focus on recent trends while still showing long-term patterns.

Visualization

The line graph below shows how the U.S. inflation rate changes over time. Each point represents the inflation rate for a given year, and the line helps show the overall trend. A red dashed line is included to represent the average inflation rate across all years. This helps compare individual years to the overall average and makes it easier to see when inflation was unusually high or low.

Table

The table below displays the exact inflation values for each year from 1995 to 2025. The values are rounded to two decimal places for readability. This provides a clear numerical reference to support the trends shown in the graph.

Conclusion

Overall, the inflation rate shows periods of both stability and fluctuation over time. While many years stay close to the average, there are noticeable spikes that likely correspond to major economic events. This analysis helps highlight how inflation has changed over time and provides a better understanding of economic trends in the United States.

U.S. Price to Income

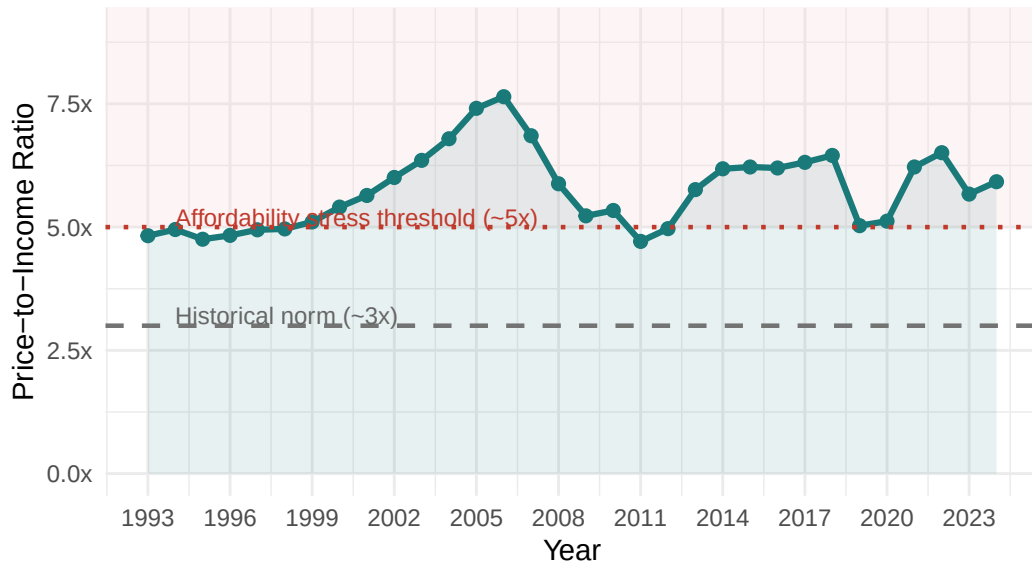
Code for Price to Income

```
Attaching package: 'scales'
```

```
The following object is masked from 'package:readr':
```

```
col_factor
```


Price-to-Income Ratio (1993–2024)



Sources: FRED — MSPUS, MEHOINUSA646N | STAT 184 Section 4

Table 3: Price-to-Income Ratio Analysis (1993–2024, showing every other year)

Median Home Price			
Year	($ MedianHouseholdIncome()$)	Price-to-Income Ratio	
1993	150750	31241	4.83
1995	162000	34076	4.75
1997	183000	37005	4.95
1999	207750	40696	5.10
2001	238250	42228	5.64
2003	275250	43318	6.35
2005	343250	46326	7.41
2007	344250	50233	6.85
2009	260250	49777	5.23
2011	235750	50054	4.71
2013	301000	52250	5.76
2015	351500	56516	6.22
2017	387500	61372	6.31
2019	345500	68703	5.03
2021	440250	70784	6.22
2023	457000	80610	5.67

Narrative Text

Introduction

The price-to-income ratio is a key measure of housing affordability that compares the median home price to the median household income. A ratio of $3\times$ is considered the historical U.S. norm, while $5\times$ is widely cited as an affordability stress threshold. By examining this ratio over time, we can understand how purchasing power for homes has changed relative to what people earn.

Data

The price-to-income analysis uses two datasets from the Federal Reserve Economic Data (FRED) system: - **MSPUS**: Median home sales price in the United States (quarterly data, averaged annually) - **MEHOINUSA646N**: Median household income in the United States (annual data)

Data Preparation

The home price data, originally provided quarterly, was aggregated into annual averages by year. Both datasets were aligned by year and then merged into a single dataframe. The price-to-income ratio was calculated as the median home price divided by the median household income. The analysis covers the period from 1993 to 2024, capturing several complete economic cycles including the housing bubble, financial crisis, and recovery period.

Visualization

The line chart shows the price-to-income ratio over the 32-year period. Two reference lines mark key affordability thresholds: a dashed line at $3\times$ representing the historical norm, and a dotted line at $5\times$ representing the stress threshold. The red-shaded region above $5\times$ indicates severe affordability challenges. The visualization reveals whether housing has become more or less affordable relative to household income over time.

Key Insights

The price-to-income ratio has **never been at or below the historical $3\times$ norm** in this dataset—it was already elevated at $4.8\times$ in 1993. The ratio peaked at $7.65\times$ in 2006 during the housing bubble, fell to $4.97\times$ in 2012 following the crisis, then climbed steadily to $6.51\times$ by 2022. By 2024, it had eased slightly to approximately $5.9\times$, remaining well above the

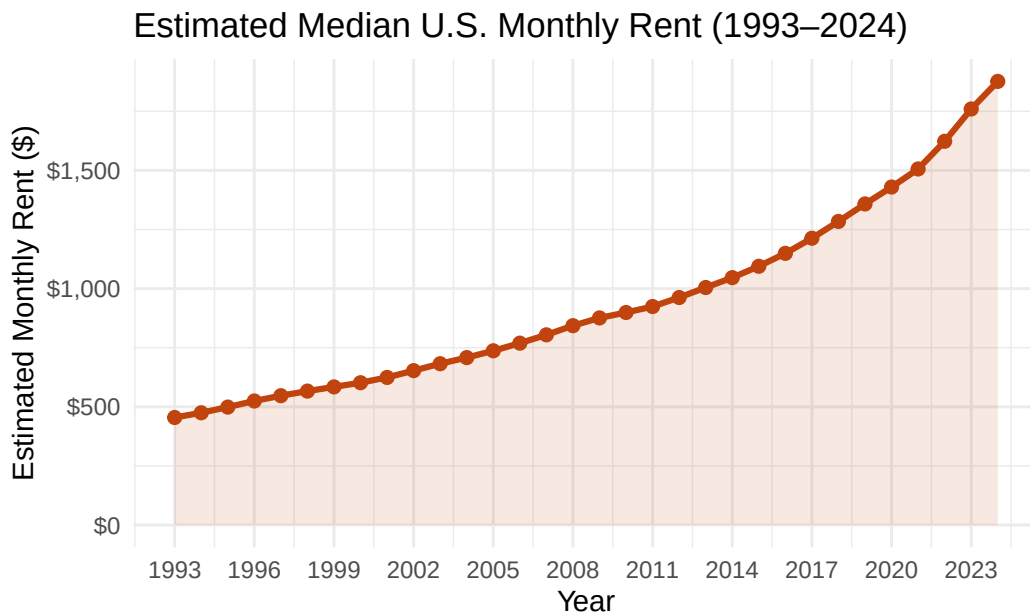
affordability stress threshold. This indicates that housing has been severely unaffordable for the median U.S. household for much of the analysis period, particularly during 2019–2024.

Conclusion

The price-to-income ratio analysis demonstrates that housing affordability for the median American household has deteriorated significantly over the past three decades. Unlike the 1990s when the ratio hovered near 4–5×, the 2020s have seen sustained ratios above 5–6×, indicating a fundamental mismatch between home prices and typical household incomes. This metric underscores the broader economic challenge of housing availability and affordability in contemporary America.

U.S. Median Rent Analysis

Code for Rent Analysis



Sources: FRED CUUR0000SEHA, U.S. Census Bureau 2000 AHS | STAT 184 Section 4

Table 4: Estimated Median Monthly Rent (1993–2024, showing every other year)

Year	Rent Index	Estimated Median Monthly Rent (\$)
1993	169.15	455.00
1995	185.50	498.98

Year	Rent Index	Estimated Median Monthly Rent (\$)
1997	203.40	547.13
1999	217.25	584.38
2001	232.10	624.33
2003	253.70	682.43
2005	274.00	737.03
2007	299.10	804.55
2009	325.70	876.10
2011	343.60	924.25
2013	373.50	1004.68
2015	406.90	1094.52
2017	451.00	1213.15
2019	504.90	1358.13
2021	559.90	1506.08
2023	654.20	1759.73

Narrative Text

Introduction

Rent is a critical component of housing affordability, as it represents the cost of renting a typical residential unit. Unlike home prices, which fluctuate significantly, rent tends to show more consistent growth patterns. Examining rent trends over three decades provides insight into how housing costs have evolved for renters in the United States.

Data

The rent analysis uses the CPI rent-of-primary-residence component (CUUR0000SEHA) from FRED, which provides a monthly index with base period 1982–84 = 100. This series is calibrated to the U.S. Census Bureau’s 2000 American Housing Survey median gross rent of \$602/month, allowing conversion from the index to nominal monthly rent values.

Data Preparation

Monthly rent index values were aggregated into annual averages. The index was then converted to nominal dollars using the 2000 Census benchmark. The analysis covers 1993 to 2024, allowing for direct comparison with home price and income trends.

Visualization

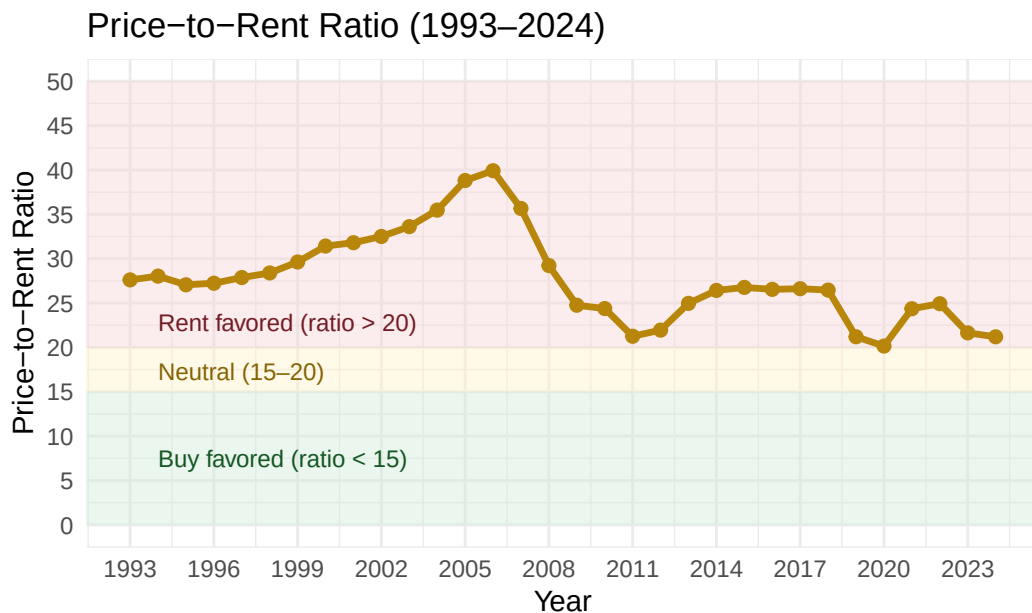
The line graph shows rent growth over the 32-year period. Unlike home prices, rent demonstrates monotonically upward growth with no significant corrections or downturns, even during economic crises.

Key Insights

From 1993 to 2024, estimated median rent rose from approximately \$455/month to \$1,876/month—a 312% increase in nominal terms. This substantially outpaced both general CPI inflation (+117%) and nominal household income growth (+158%) over the same period. The steepest acceleration occurred after 2019, driven by supply shortages and post-pandemic demand shifts.

U.S. Price-to-Rent Ratio

Code for Price-to-Rent Ratio



Sources: FRED MSPUS, CUUR0000SEHA | STAT 184 Section 4

Table 5: Price-to-Rent Ratio Analysis (1993–2024, showing every other year)

Year	Median Home Price ()	$AnnualRent()$	Price-to-Rent Ratio
1993	150750	5459.962	27.6
1995	162000	5987.721	27.1
1997	183000	6565.512	27.9
1999	207750	7012.574	29.6
2001	238250	7491.914	31.8
2003	275250	8189.137	33.6
2005	343250	8844.397	38.8
2007	344250	9654.595	35.7
2009	260250	10513.212	24.8
2011	235750	11091.003	21.3
2013	301000	12056.139	25.0
2015	351500	13134.252	26.8
2017	387500	14557.748	26.6
2019	345500	16297.576	21.2
2021	440250	18072.912	24.4
2023	457000	21116.804	21.6

Narrative Text

Introduction

The price-to-rent ratio compares the annual cost of owning (home price) to the annual cost of renting (annual rent). This ratio helps determine whether it is financially advantageous to buy or rent a property. A ratio below 15 traditionally favors buying, 15–20 is considered neutral, and above 20 favors renting.

Data

The price-to-rent analysis combines median home sales prices (MSPUS) with the CPI rent index (CUUR0000SEHA), calibrated to the 2000 Census benchmark.

Data Preparation

The annual rent is calculated by multiplying the monthly rent estimate by 12. The price-to-rent ratio is then computed as the median home price divided by annual rent.

Key Insights

The price-to-rent ratio has remained entirely in “rent-favored” territory (above 20×) throughout the 1993–2024 window, ranging from 20.2 to 39.9. It peaked during the mid-2000s bubble (~40×), fell post-crisis to ~22×, recovered through the 2010s, then spiked again during the 2020–2022 pandemic surge before easing to ~21× by 2024. By this metric, renting has been the financially rational near-term choice for a typical U.S. household for over three decades.

Summary and Key Findings

Table 6: Summary of Key Metrics: 1993 vs. 2024

Metric	X1993	X2024	Change..
Median Home Price	\$151,000	\$477,000	217%
Median Rent	\$455/month	\$1,876/month	312%
Nominal Income Growth	—	\$80,610	158%
Real Income Growth	—	\$80,610 (2024 \$)	19%
Price-to-Income Ratio	4.8x	5.9x	+23%
Price-to-Rent Ratio	20.2x	21.0x	+4%
CPI Inflation	144.5	313.7	117%

Key Findings

- Housing Affordability Crisis:** Median home prices rose 217% between 1993 and 2024, while real (inflation-adjusted) household income grew only 19%. The price-to-income ratio has remained above 5× since 2019, indicating severe affordability stress for the median U.S. household.
- Rent Acceleration:** Estimated median rent surged 312% over the same period—faster than home prices—driven by supply shortages and post-pandemic demand. Monthly rent has grown from \$455 to nearly \$1,900, substantially exceeding income growth.
- Price-to-Rent Ratio:** With ratios consistently above 20×, renting has been financially advantageous compared to buying for over three decades. The 2020–2022 period saw particular pressure, with ratios peaking around 25–30×.
- Inflation and Real Income:** While nominal incomes rose 158%, inflation of 117% over 31 years meant real income gains were minimal. The 2021–2022 inflation surge (+7.96% annual rate) caused the sharpest real income decline since the early 1980s.

5. **Long-term Trends:** The analysis reveals two major inflection points: the 2006–2011 housing crisis (prices fell ~35% from peak) and the 2020–2022 pandemic surge (fastest price growth in the dataset). Both events demonstrate housing market volatility and its disconnect from wage growth.

Conclusion

The 30-year analysis of U.S. housing affordability reveals a fundamental and worsening mismatch between home prices, rents, and household incomes. While nominal metrics show substantial growth across all categories, real income growth has been negligible, eroded by persistent inflation. The price-to-income ratio, consistently above the $5\times$ stress threshold since 2019, and the price-to-rent ratio, above the “rent-favored” threshold throughout the entire study period, both indicate severe housing affordability challenges for median American households. These findings underscore the critical importance of housing policy interventions and income growth strategies in addressing the affordability crisis in contemporary America.